

Differentiation

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January 2014 Q1 C34

1.
$$f(x) = \frac{2x}{x^2 + 3}, \quad x \in \mathbb{R}$$

Find the set of values of x for which $f'(x) > 0$

You must show your working.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(6)

June 2014 Q3 C34

3. Given that

$$y = \frac{\cos 2\theta}{1 + \sin 2\theta}, \quad -\frac{\pi}{4} < \theta < \frac{3\pi}{4}$$

show that

$$\frac{dy}{d\theta} = \frac{a}{1 + \sin 2\theta}, \quad -\frac{\pi}{4} < \theta < \frac{3\pi}{4}$$

where a is a constant to be determined.

(4)

June 2014 Q10 C34

10. (a) Use the identity for $\sin(A + B)$ to prove that

$$\sin 2A \equiv 2 \sin A \cos A \quad (2)$$

- (b) Show that

$$\frac{d}{dx} \left[\ln \left(\tan \left(\frac{1}{2} x \right) \right) \right] = \operatorname{cosec} x \quad (4)$$

A curve C has the equation

$$y = \ln \left(\tan \left(\frac{1}{2} x \right) \right) - 3 \sin x, \quad 0 < x < \pi$$

- (c) Find the x coordinates of the points on C where $\frac{dy}{dx} = 0$

Give your answers to 3 decimal places.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(6)

January 2015 Q1 C34

1. The curve C has equation

$$y = \frac{3x - 2}{(x - 2)^2}, \quad x \neq 2$$

The point P on C has x coordinate 3

Find an equation of the normal to C at the point P in the form $ax + by + c = 0$, where a , b and c are integers.

(6)

3.

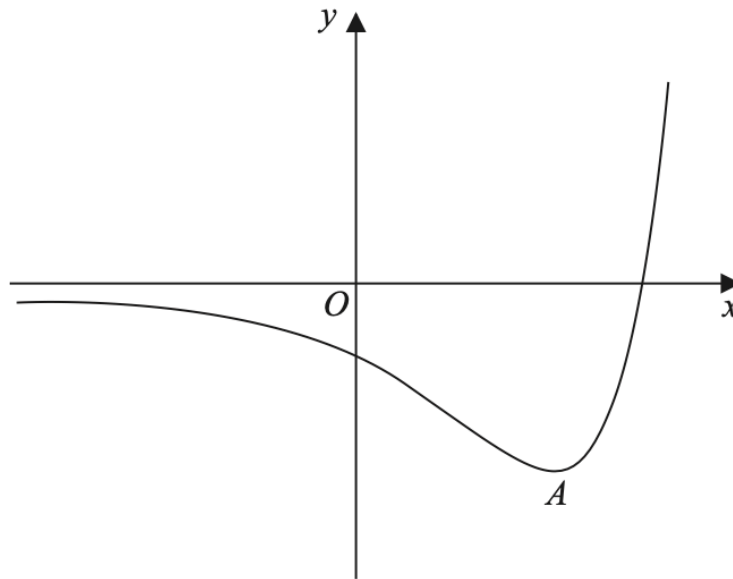


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = (2x - 5)e^x, \quad x \in \mathbb{R}$$

The curve has a minimum turning point at A .

(a) Use calculus to find the exact coordinates of A .

(5)

Given that the equation $f(x) = k$, where k is a constant, has exactly two roots,

(b) state the range of possible values of k .

(2)

(c) Sketch the curve with equation $y = |f(x)|$.

Indicate clearly on your sketch the coordinates of the points at which the curve crosses or meets the axes.

(3)

7.

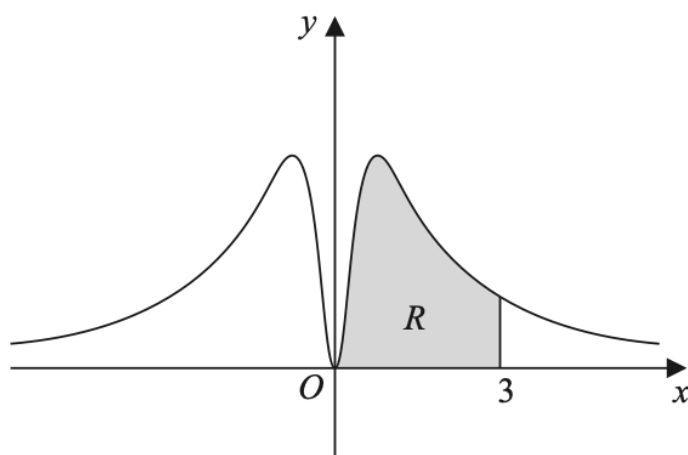


Figure 3

Figure 3 shows part of the curve C with equation

$$y = \frac{3 \ln(x^2 + 1)}{(x^2 + 1)}, \quad x \in \mathbb{R}$$

- (a) Find $\frac{dy}{dx}$ (2)
- (b) Using your answer to (a), find the exact coordinates of the stationary point on the curve C for which $x > 0$. Write each coordinate in its simplest form. (5)

The finite region R , shown shaded in Figure 3, is bounded by the curve C , the x -axis and the line $x = 3$

- (c) Complete the table below with the value of y corresponding to $x = 1$

x	0	1	2	3
y	0		$\frac{3}{5} \ln 5$	$\frac{3}{10} \ln 10$

(1)

- (d) Use the trapezium rule with all the y values in the completed table to find an approximate value for the area of R , giving your answer to 4 significant figures. (3)

January 2017 Q6 C34

6. (i) Differentiate $y = 5x^2 \ln 3x$, $x > 0$

(2)

- (ii) Given that

$$y = \frac{x}{\sin x + \cos x}, \quad -\frac{\pi}{4} < x < \frac{3\pi}{4}$$

show that

$$\frac{dy}{dx} = \frac{(1+x)\sin x + (1-x)\cos x}{1 + \sin 2x}, \quad -\frac{\pi}{4} < x < \frac{3\pi}{4}$$

(4)

June 2017 Q5 C34

5.

$$\frac{6 - 5x - 4x^2}{(2 - x)(1 + 2x)} \equiv A + \frac{B}{(2 - x)} + \frac{C}{(1 + 2x)}$$

(a) Find the values of the constants A , B and C .

(4)

$$f(x) = \frac{6 - 5x - 4x^2}{(2 - x)(1 + 2x)} \quad x > 2$$

(b) Using part (a), find $f'(x)$.

(3)

(c) Prove that $f(x)$ is a decreasing function.

(1)

June 2017 Q11 C34

11.
$$y = (2x^2 - 3) \tan\left(\frac{1}{2}x\right), \quad 0 < x < \pi$$

(a) Find the exact value of x when $y = 0$

(1)

Given that $\frac{dy}{dx} = 0$ when $x = \alpha$,

(b) show that

$$2\alpha^2 - 3 + 4\alpha \sin \alpha = 0$$

(6)

The iterative formula

$$x_{n+1} = \frac{3}{(2x_n + 4 \sin x_n)}$$

can be used to find an approximation for α .

(c) Taking $x_1 = 0.7$, find the values of x_2 and x_3 , giving each answer to 4 decimal places.

(2)

(d) By choosing a suitable interval, show that $\alpha = 0.7283$ to 4 decimal places.

(2)

6.

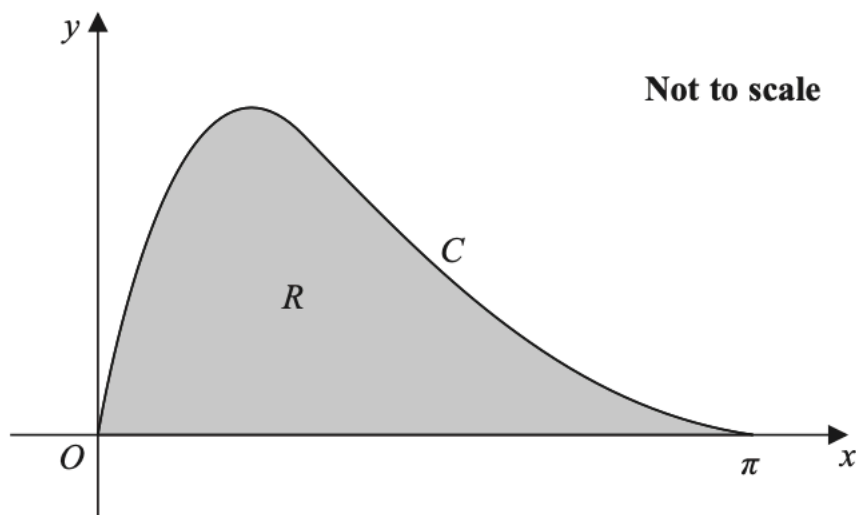


Figure 1

Figure 1 shows a sketch of the curve C with equation $y = 2e^{-x}\sqrt{\sin x}$, $0 \leq x \leq \pi$. The finite region R , shown shaded in Figure 1, is bounded by the curve and the x -axis.

- (a) Complete the table below with the value of y corresponding to $x = \frac{\pi}{2}$, giving your answer to 5 decimal places.

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
y	0	0.76679		0.15940	0

(1)

- (b) Use the trapezium rule, with all the values of y in the completed table, to obtain an estimate for the area of the region R . Give your answer to 4 decimal places.

(3)

- (c) Given $y = 2e^{-x}\sqrt{\sin x}$, find $\frac{dy}{dx}$ for $0 < x < \pi$.

(3)

The curve C has a maximum turning point when $x = a$.

- (d) Use your answer to part (c) to find the value of a , giving your answer to 3 decimal places.

(3)

June 2018 Q14 C34

14. Given that

$$y = \frac{(x^2 - 4)^{\frac{1}{2}}}{x^3} \quad x > 2$$

(a) show that

$$\frac{dy}{dx} = \frac{Ax^2 + 12}{x^4(x^2 - 4)^{\frac{1}{2}}} \quad x > 2$$

where A is a constant to be found.

(6)

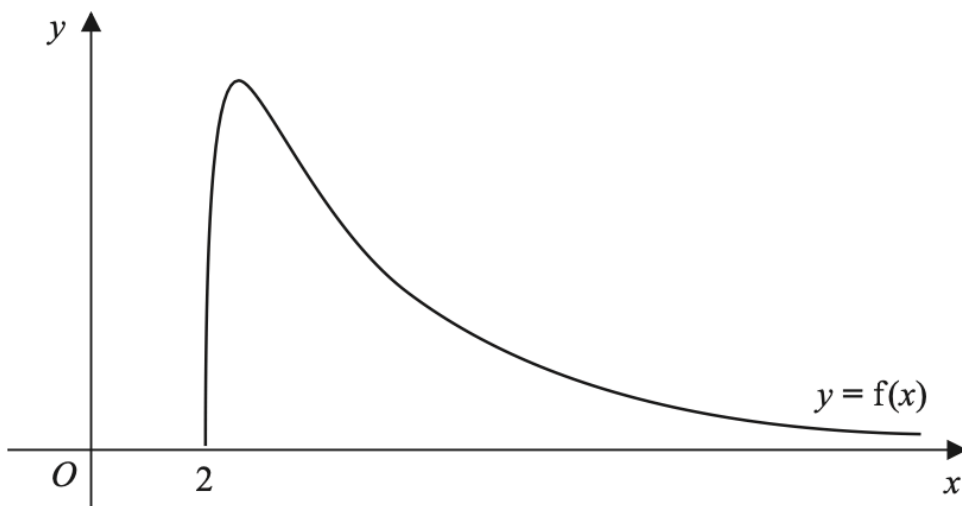


Figure 4

Figure 4 shows a sketch of part of the curve with equation $y = f(x)$ where

$$f(x) = \frac{24(x^2 - 4)^{\frac{1}{2}}}{x^3} \quad x > 2$$

(b) Use your answer to part (a) to find the range of f .

(5)

(c) State a reason why f^{-1} does not exist.

(1)

October 2018 Q7 C34

7. A curve has equation

$$y = \ln(1 - \cos 2x), \quad x \in \mathbb{R}, 0 < x < \pi$$

Show that

(a) $\frac{dy}{dx} = k \cot x$, where k is a constant to be found. (4)

Hence find the exact coordinates of the point on the curve where

(b) $\frac{dy}{dx} = 2\sqrt{3}$ (4)

8.

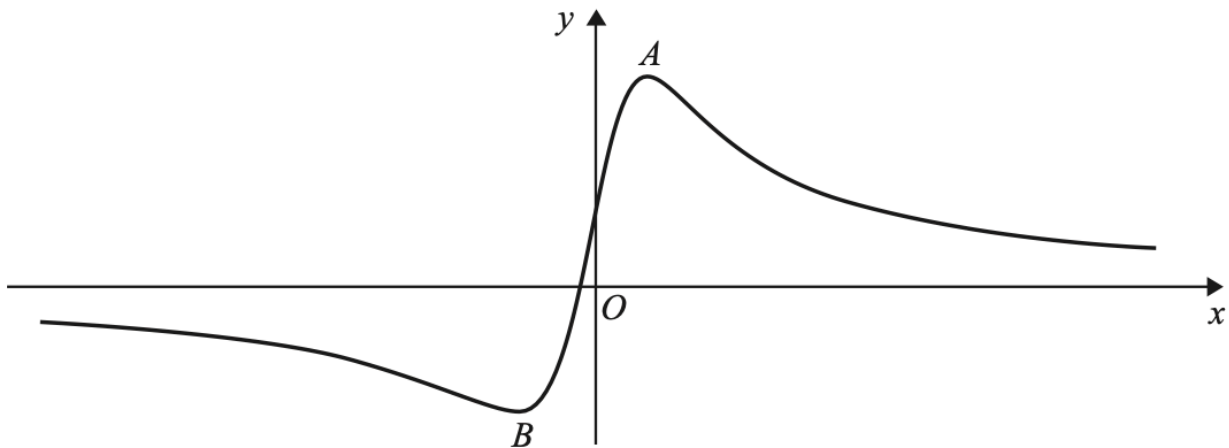


Figure 2

Figure 2 shows a sketch of part of the curve $y = f(x)$, where

$$f(x) = \frac{6x + 2}{3x^2 + 5}, \quad x \in \mathbb{R}$$

- (a) Find $f'(x)$, writing your answer as a single fraction in its simplest form. (3)

The curve has two turning points, a maximum at point A and a minimum at point B , as shown in Figure 2.

- (b) Using part (a), find the coordinates of point A and the coordinates of point B . (4)

- (c) State the coordinates of the maximum turning point of the function with equation

$$y = f(2x) + 4 \quad x \in \mathbb{R} \quad (2)$$

- (d) Find the range of the function

$$g(x) = \frac{6x + 2}{3x^2 + 5}, \quad x \leq 0 \quad (2)$$

January 2020 Q4 P3

4. (i)

$$f(x) = \frac{(2x + 5)^2}{x - 3} \quad x \neq 3$$

(a) Find $f'(x)$ in the form $\frac{P(x)}{Q(x)}$ where $P(x)$ and $Q(x)$ are fully factorised quadratic expressions.

(b) Hence find the range of values of x for which $f(x)$ is increasing.

(6)

(ii)

$$g(x) = x\sqrt{\sin 4x} \quad 0 \leq x < \frac{\pi}{4}$$

The curve with equation $y = g(x)$ has a maximum at the point M .

Show that the x coordinate of M satisfies the equation

$$\tan 4x + kx = 0$$

where k is a constant to be found.

(5)

3.

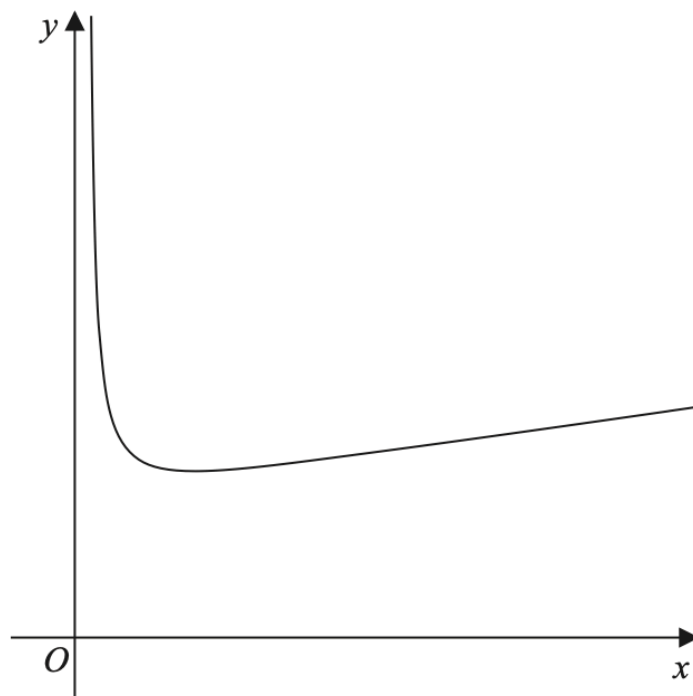


Figure 1

Figure 1 shows a sketch of a curve with equation $y = f(x)$ where

$$f(x) = \frac{2x + 3}{\sqrt{4x - 1}} \quad x > \frac{1}{4}$$

(a) Find, in simplest form, $f'(x)$.

(4)

(b) Hence find the range of f .

(3)

October 2020 Q8 P3

8. (i) The curve C has equation $y = g(x)$ where

$$g(x) = e^{3x} \sec 2x \quad -\frac{\pi}{4} < x < \frac{\pi}{4}$$

- (a) Find $g'(x)$

(2)

- (b) Hence find the x coordinate of the stationary point of C .

(3)

- (ii) A different curve has equation

$$x = \ln(\sin y) \quad 0 < y < \frac{\pi}{2}$$

Show that

$$\frac{dy}{dx} = \frac{e^x}{f(x)}$$

where $f(x)$ is a function of e^x that should be found.

(4)

January 2021 Q10 P3

10. The curve C has equation

$$x = 3\sec^2 2y \quad x > 3 \quad 0 < y < \frac{\pi}{4}$$

(a) Find $\frac{dx}{dy}$ in terms of y .

(2)

(b) Hence show that

$$\frac{dy}{dx} = \frac{p}{qx\sqrt{x-3}}$$

where p is irrational and q is an integer, stating the values of p and q .

(3)

(c) Find the equation of the normal to C at the point where $y = \frac{\pi}{12}$, giving your answer in the form $y = mx + c$, giving m and c as exact irrational numbers.

(5)

June 2021 Q1 P3

1. The curve C has equation

$$y = x^2 \cos\left(\frac{1}{2}x\right) \quad 0 < x \leq \pi$$

The curve has a stationary point at the point P .

- (a) Show, using calculus, that the x coordinate of P is a solution of the equation

$$x = 2 \arctan\left(\frac{4}{x}\right) \quad (4)$$

Using the iteration formula

$$x_{n+1} = 2 \arctan\left(\frac{4}{x_n}\right) \quad x_1 = 2$$

- (b) find the value of x_2 and the value of x_6 , giving your answers to 3 decimal places.

(3)

June 2021 Q7 P3

7. Given that

$$x = 6 \sin^2 2y \qquad 0 < y < \frac{\pi}{4}$$

show that

$$\frac{dy}{dx} = \frac{1}{A\sqrt{Bx - x^2}}$$

where A and B are integers to be found.

(5)

October 2021 Q1 P3

1. The function f is defined by

$$f(x) = \frac{5x}{x^2 + 7x + 12} + \frac{5x}{x + 4} \quad x > 0$$

(a) Show that $f(x) = \frac{5x}{x + 3}$ (3)

(b) Find f^{-1} (3)

(c) (i) Find, in simplest form, $f'(x)$.

(ii) Hence, state whether f is an increasing or a decreasing function, giving a reason for your answer. (3)

October 2021 Q6 P3

6. (i) The curve C_1 has equation

$$y = 3 \ln(x^2 - 5) - 4x^2 + 15 \quad x > \sqrt{5}$$

Show that C_1 has a stationary point at $x = \frac{\sqrt{p}}{2}$ where p is a constant to be found. (4)

- (ii) A different curve C_2 has equation

$$y = 4x - 12 \sin^2 x$$

- (a) Show that, for this curve,

$$\frac{dy}{dx} = A + B \sin 2x$$

where A and B are constants to be found.

- (b) Hence, state the maximum gradient of this curve.

(4)

October 2021 Q8 P3

8. A curve C has equation $y = f(x)$, where

$$f(x) = \arcsin\left(\frac{1}{2}x\right) \quad -2 \leq x \leq 2 \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

(a) Sketch C .

(1)

(b) Given $x = 2 \sin y$, show that

$$\frac{dy}{dx} = \frac{1}{\sqrt{A - x^2}}$$

where A is a constant to be found.

(3)

The point P lies on C and has y coordinate $\frac{\pi}{4}$

(c) Find the equation of the tangent to C at P . Write your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

January 2022 Q1 P3

1. Find, using calculus, the x coordinate of the stationary point on the curve with equation

$$y = (2x + 5)e^{3x}$$

(4)

5.

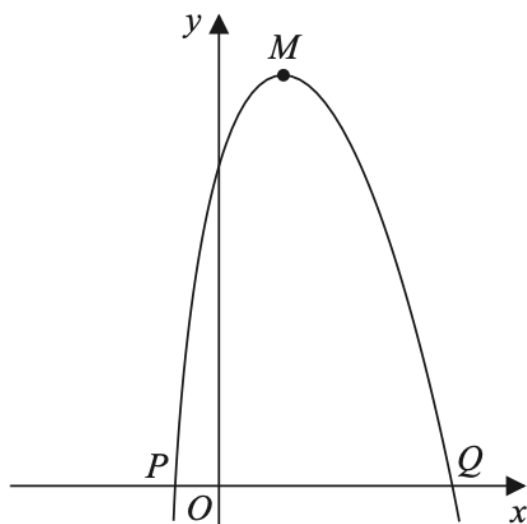


Figure 1

Figure 1 shows a sketch of part of the curve with equation

$$y = 6 \ln(2x + 3) - \frac{1}{2}x^2 + 4 \quad x > -\frac{3}{2}$$

The curve cuts the negative x -axis at the point P , as shown in Figure 1.

(a) Show that the x coordinate of P lies in the interval $[-1.25, -1.2]$

(2)

The curve cuts the positive x -axis at the point Q , also shown in Figure 1.

Using the iterative formula

$$x_{n+1} = \sqrt{12 \ln(2x_n + 3) + 8} \quad \text{with } x_1 = 6$$

(b) (i) find, to 4 decimal places, the value of x_2

(ii) find, by continued iteration, the x coordinate of Q . Give your answer to 4 decimal places.

(3)

The curve has a maximum turning point at M , as shown in Figure 1.

(c) Using calculus and showing each stage of your working, find the x coordinate of M .

(4)

January 2022 Q6 P3

6. The function f is defined by

$$f(x) = \frac{5x - 3}{x - 4} \quad x > 4$$

(a) Show, by using calculus, that f is a decreasing function.

(3)

(b) Find f^{-1}

(3)

(c) (i) Show that $ff(x) = \frac{ax + b}{x + c}$ where a , b and c are constants to be found.

(ii) Deduce the range of ff .

(5)

10.

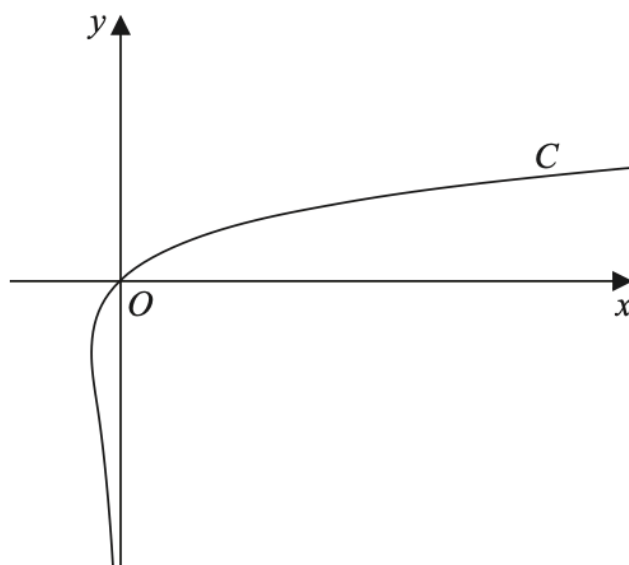


Figure 3

Figure 3 shows a sketch of the curve C with equation

$$x = ye^{2y} \quad y \in \mathbb{R}$$

(a) Show that

$$\frac{dy}{dx} = \frac{y}{x(1+2y)} \quad (4)$$

Given that the straight line with equation $x = k$, where k is a constant, cuts C at exactly two points,

(b) find the range of possible values for k .

(3)

June 2022 Q1 P3

1. The curve C has equation

$$y = (3x - 2)^6$$

- (a) Find $\frac{dy}{dx}$

(2)

Given that the point $P\left(\frac{1}{3}, 1\right)$ lies on C ,

- (b) find the equation of the normal to C at P . Write your answer in the form $ax + by + c = 0$ where a , b and c are integers to be found.

(4)

6.

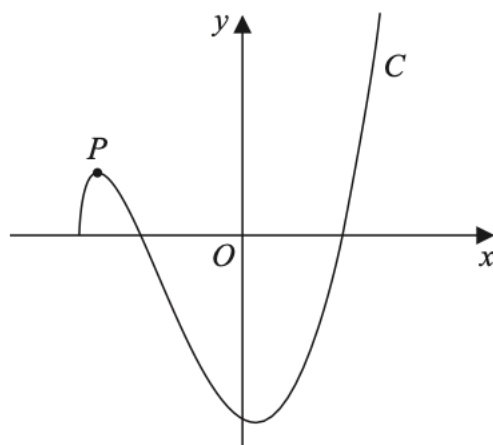


Figure 3

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The function f is defined by

$$f(x) = 5(x^2 - 2)(4x + 9)^{\frac{1}{2}} \quad x \geq -\frac{9}{4}$$

(a) Show that

$$f'(x) = \frac{k(5x^2 + 9x - 2)}{(4x + 9)^{\frac{1}{2}}}$$

where k is an integer to be found.

(4)

(b) Hence, find the values of x for which $f'(x) = 0$

(1)

Figure 3 shows a sketch of the curve C with equation $y = f(x)$.

The curve has a local maximum at the point P

(c) Find the exact coordinates of P

(2)

The function g is defined by

$$g(x) = 2f(x) + 4 \quad -\frac{9}{4} \leq x \leq 0$$

(d) Find the range of g

(3)

9.

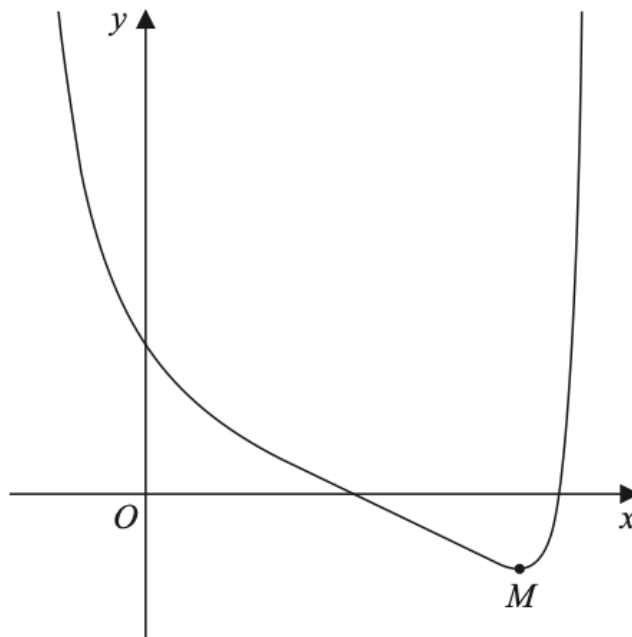


Figure 5

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Figure 5 shows the curve with equation

$$y = \frac{1 + 2 \cos x}{1 + \sin x} \quad -\frac{\pi}{2} < x < \frac{3\pi}{2}$$

The point M , shown in Figure 5, is the minimum point on the curve.

(a) Show that the x coordinate of M is a solution of the equation

$$2 \sin x + \cos x = -2$$

(4)

(b) Hence find, to 3 significant figures, the x coordinate of M .

(5)

October 2022 Q3 P3

3.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

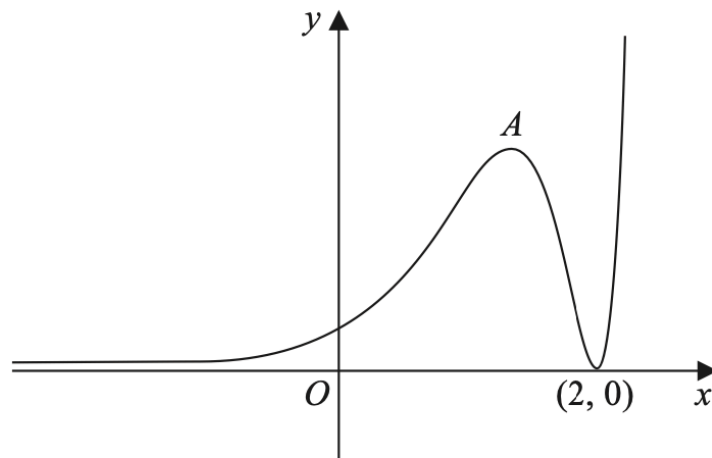


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = f(x)$ where

$$f(x) = (x - 2)^2 e^{3x} \quad x \in \mathbb{R}$$

The curve has a maximum turning point at A and a minimum turning point at $(2, 0)$

(a) Use calculus to find the exact coordinates of A .

(5)

Given that the equation $f(x) = k$, where k is a constant, has **at least** two distinct roots,

(b) state the range of possible values for k .

(2)

October 2022 Q6 P3

6.

$$y = \frac{2 + 3 \sin x}{\cos x + \sin x}$$

Show that

$$\frac{dy}{dx} = \frac{a \tan x + b \sec x + c}{\sec x + 2 \sin x}$$

where a , b and c are integers to be found.

(6)

January 2023 Q7 P3

7. The curve C has equation

$$x = 3 \tan\left(y - \frac{\pi}{6}\right) \quad x \in \mathbb{R} \quad -\frac{\pi}{3} < y < \frac{2\pi}{3}$$

(a) Show that

$$\frac{dy}{dx} = \frac{a}{x^2 + b}$$

where a and b are integers to be found.

(4)

The point P with y coordinate $\frac{\pi}{3}$ lies on C .

Given that the tangent to C at P crosses the x -axis at the point Q .

(b) find, in simplest form, the exact x coordinate of Q .

(5)

8.

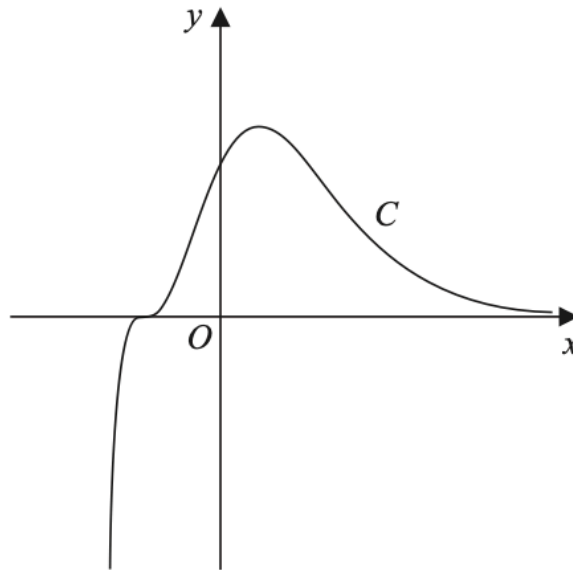


Figure 3

Figure 3 shows a sketch of the curve C with equation $y = f(x)$, where

$$f(x) = (2x + 1)^3 e^{-4x}$$

(a) Show that

$$f'(x) = A(2x + 1)^2 (1 - 4x) e^{-4x}$$

where A is a constant to be found.

(4)

(b) Hence find the exact coordinates of the two stationary points on C .

(3)

The function g is defined by

$$g(x) = 8f(x - 2)$$

(c) Find the coordinates of the maximum stationary point on the curve with equation $y = g(x)$.

(2)

10.

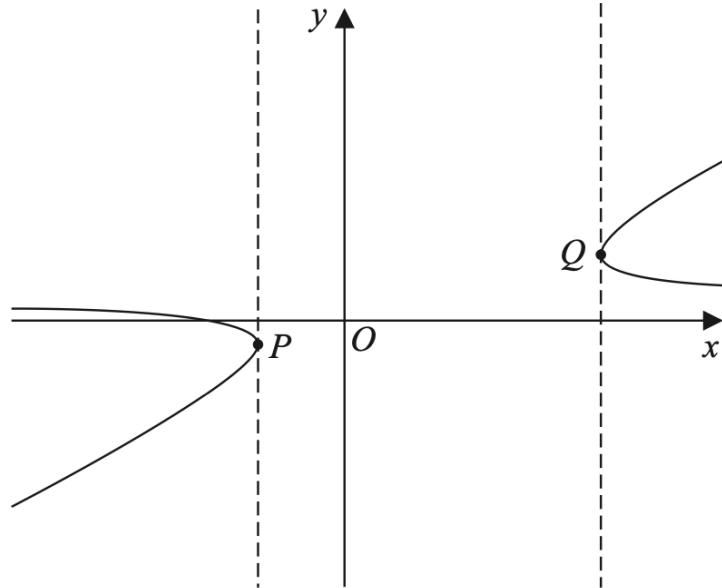


Figure 4

Figure 4 shows a sketch of the curve with equation

$$x = \frac{2y^2 + 6}{3y - 3}$$

- (a) Find $\frac{dx}{dy}$ giving your answer as a fully simplified fraction.

(4)

The tangents at points P and Q on the curve are parallel to the y -axis, as shown in Figure 4.

- (b) Use the answer to part (a) to find the equations of these two tangents.

(4)

October 2023 Q5 P3

5. The curve C has equation

$$y = \frac{\ln(x^2 + k)}{x^2 + k} \quad x \in \mathbb{R}$$

where k is a positive constant.

(a) Show that

$$\frac{dy}{dx} = \frac{Ax(B - \ln(x^2 + k))}{(x^2 + k)^2}$$

where A and B are constants to be found.

(3)

Given that C has exactly three turning points,

(b) find the x coordinate of each of these points. Give your answer in terms of k where appropriate.

(3)

(c) find the upper limit to the value for k .

(1)

7.

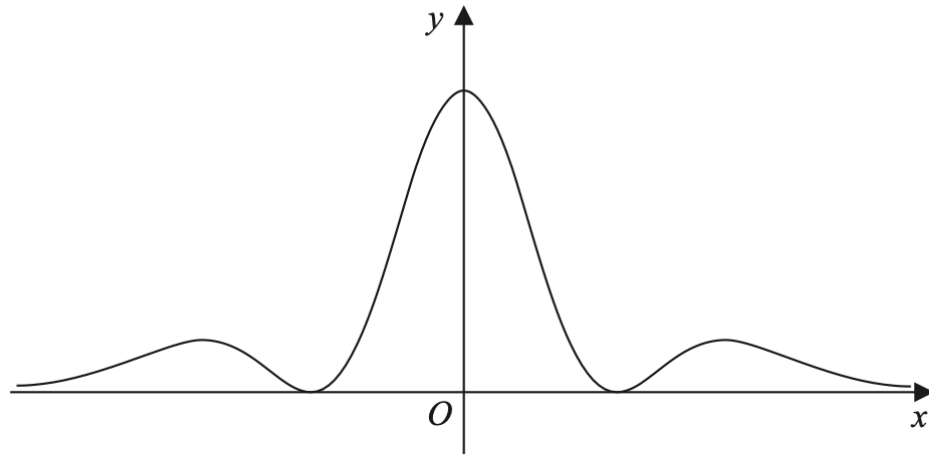


Figure 1

Figure 1 shows a sketch of the curve C with equation $y = f(x)$ where

$$f(x) = e^{-x^2} (2x^2 - 3)^2$$

(a) Find the range of f

(2)

(b) Show that

$$f'(x) = 2x(2x^2 - 3)e^{-x^2}(A - Bx^2)$$

where A and B are constants to be found.

(4)

Given that the line $y = k$, where k is a constant, $k > 0$, intersects the curve at exactly two distinct points,

(c) find the exact range of values of k

(4)

October 2023 Q10 P3

10.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve C has equation

$$x = \sin^2 4y \quad 0 \leq y \leq \frac{\pi}{8} \quad 0 \leq x \leq 1$$

The point P with x coordinate $\frac{1}{4}$ lies on C

(a) Find the exact y coordinate of P (2)

(b) Find $\frac{dx}{dy}$ (2)

(c) Hence show that $\frac{dy}{dx}$ can be written in the form

$$\frac{dy}{dx} = \frac{1}{\sqrt{q + r(x + s)^2}}$$

where q , r and s are constants to be found.

(3)

Using the answer to part (c),

(d) (i) state the x coordinate of the point where the value of $\frac{dy}{dx}$ is a minimum,

(ii) state the value of $\frac{dy}{dx}$ at this point.

(2)

6.

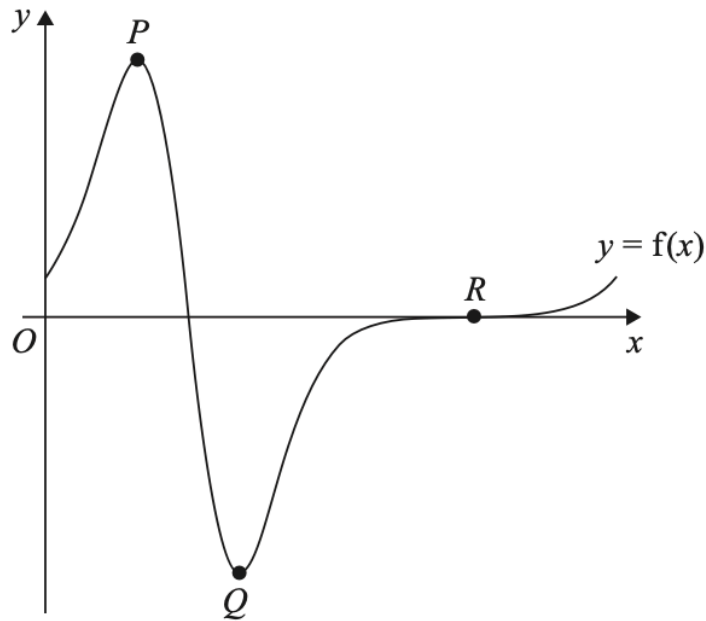


Figure 1

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Figure 1 shows a sketch of the curve with equation $y = f(x)$, where

$$f(x) = 2e^{3\sin x} \cos x \quad 0 \leq x \leq 2\pi$$

The curve intersects the x -axis at point R , as shown in Figure 1.

(a) State the coordinates of R

(1)

The curve has two turning points, at point P and point Q , also shown in Figure 1.

(b) Show that, at points P and Q ,

$$a \sin^2 x + b \sin x + c = 0$$

where a , b and c are integers to be found.

(4)

(c) Hence find the x coordinate of point Q , giving your answer to 3 decimal places.

(2)

January 2024 Q7 P3

7. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = \frac{16}{9(3x - k)} \quad x \neq \frac{k}{3}$$

where k is a positive constant not equal to 3

(a) Find $\frac{dy}{dx}$ giving your answer in simplest form in terms of k . (2)

The point P with x coordinate 1 lies on C .

Given that the gradient of the curve at P is -12

(b) find the two possible values of k (3)

Given also that $k < 3$

(c) find the equation of the normal to C at P , writing your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found. (3)

(d) show, using algebraic integration that,

$$\int_1^3 \frac{16}{9(3x - k)} dx = \lambda \ln 10$$

where λ is a constant to be found.

(4)